Protocol Setting

General Information		
Baud Rate		9600
Parity		None
Data Bits		8
Stop Bits		1
Addresses Mode		PLC Addresses Mode (Base 1)

ADU (Application Data Unit)			
	PDU (Protocol Data Unit)		
Address	Function Code	Data	Error Check
1 byte	1 bytes	<=252 byte	2 bytes
Address		Inverter ID=1~255 **(Default : 5)**	
Function Code		0x01: Read Coils	
		0x02: Read Discrete Input	
		0x03: Read Holding Register	
		0x04: Read Input Register	
		0x05: Write Single Coil	
		0x06: Write Single Single Register	
		0x10: Write Multiple Registers	
Data		Refer to the Modbus Specification V1.1	
Error Check		CRC	

	Address (DHex)	Address (Dec)	Description	
	400	1024	*reserved*	
	401	1025	*reserved*	
	402	1026	*reserved*	
	403	1027	*reserved*	
	404	1028	*reserved*	
	405	1029	*reserved*	
	406	1030	*reserved*	
	407	1031	*reserved*	
0	408	1032	*Inverter ID*	
	409	1033	*RTC_Year*	
	40A	1034	*RTC_Month*	
	40B	1035	*RTC_Day*	
	40C	1036	*RTC_Hour*	
	40D	1037	*RTC_Minute*	
	40E	1038	*RTC_Second*	
	40F	1039	*reserved*	
	410	1040	*FW1(DSP) Revision*	*ex : 0x0A0B => V10.11*
	411	1041	*FW1 Date*	*ex : 0x082D => Y08W45*
	412	1042	*FW2(RED) Revision*	*ex : 0x0A0B => V10.11*
	413	1043	*FW2 Date*	*ex : 0x082D => Y08W45*
	414	1044	*FW3(COM) Revision*	*ex : 0x0A0B => V10.11*
	415	1045	*FW3 Date*	*ex : 0x082D => Y08W45*
	416	1046	*FW4 Revision*	*ex : 0x0A0B => V10.11*
	417	1047	*FW4 Date*	*ex : 0x082D => Y08W45*
1	418	1048	*Inverter Status*	
	419	1049	*Reconnected Time*	*unit : second*
	41A	1050	*Device_Type*	*unit : 100W ; ex : 200 => 20KW, 150 => 15KW, 0 => default(searching)*
	41B	1051	*CheckEvenState*	*for ATS use, PASS : 0x1234, Fail : 0x5A5A, NoT Ready : 0x0000*
	41C	1052	*CheckProtocolState*	*for ATS use, PASS : 0x1234, Fail : 0x5A5A*
	41D	1053	*Software Country List ID*	*0x0000 : Not available* *bit 15~12: 1- 3phase model* *2- 1phase model* *bit 11~0: 0 - Delta*
	41E	1054	*Adc HW power rating 2 -- ATS*	*Delta - IR 1054 and HR 33597*
	41F	1055	*Adc HW power rating 1-- ATS*	
	420	1056	Measurement Index	*HoldingRegisters:800*
	421	1057	*Voltage*	*unit : 0.1V*
	422	1058	*Current*	*unit : 0.01A*
	423	1059	*Wattage*	*unit : 1W*
	424	1060	*Frequency*	*unit : 0.01Hz*
	425	1061	*Percentage*	*unit : 1%*
	426	1062	*Redundant Voltage*	*unit : 0.1V*

2	427	1063	*Redundant Frequency*	unit : 0.01Hz
	428	1064	*Dc Offset Compensation*	unit : 0.01Hz
	429	1065	*Dc Offset Reference*	unit : 1 ADC count
	42A	1066	*Apparent Power*	
	42B	1067	*Adc Apparent Power*	
	42C	1068	*Adc Voltage*	unit : 1 ADC count
	42D	1069	*Adc Current*	unit : 1 ADC count
	42E	1070	*Adc Wattage*	unit : 1 ADC count
	42F	1071	*Adc Redundant Voltage*	unit : 1 ADC count
3	430	1072	*Today Wh*	1072:Low Word, 1073: High Word unit : 10Wh
	431	1073		
	432	1074	*Today Runtime*	1074:Low Word, 1075: High Word unit : 1 second
	433	1075		
	434	1076	*Life Wh*	1076:Low Word, 1077: High Word unit : 10Wh
	435	1077		
	436	1078	*Life Runtime*	1078:Low Word, 1079: High Word unit : 1 second
	437	1079		
	438	1080	*Temperature 1*	unit : 1C
	439	1081	*Temperature 2*	unit : 1C
	43A	1082	*Temperature 3*	unit : 1C
	43B	1083	*Temperature 4*	unit : 1C
	43C	1084	*Temperature 5*	unit : 1C
	43D	1085	*Temperature 6*	unit : 1C
	43E	1086	*Temperature 7*	unit : 1C
	43F	1087	*Temperature 8*	unit : 1C
4	440	1088	Event Index	This shall be assigned via Holding Registers(801) first when reading the events. The event index is the page index. For current design in PV inverter the maximum event is 30, so the index is from 0 to 2. If the index is equal to 0, the event with index from 0 to 9 will be reported. If the index is equal to 1, the event with index from 10 to 19 will be reported, and so on.
	441	1089	*Event (index+0)*	
	442	1090	*Event (index+1)*	
	443	1091	*Event (index+2)*	
	444	1092	*Event (index+3)*	
	445	1093	*Event (index+4)*	
	446	1094	*Event (index+5)*	
	447	1095	*Event (index+6)*	
	448	1096	*Event (index+7)*	
	449	1097	*Event (index+8)*	
	44A	1098	*Event (index+9)*	
	44B	1099	*reserved*	
	44C	1100	*reserved*	
	44D	1101	*reserved*	
	44E	1102	*reserved*	
	44F	1103	*reserved*	
	450	1104	*Inter RS485 Quality*	
	451	1105	*History Even Count*	

TL5k: Amb Temp
TL5k: BoostHs Temp
TL5k: InvHs Temp

5	452	1106	*Derating(for ATS)* *- bit 0 : Thermal Derating* *- bit 1 - 15 :Reserved*	
	453	1107	*calculate DC on counter (for DQ testing)*	
	454	1108	*reserved*	
	455	1109	*Delta FW1(DSP) Revision*	*ex : 0x0A0B => V10.11*
	456	1110	*Delta FW2(RED) Revision*	
	457	1111	*Delta FW3(COM) Revision*	
	458	1112	*Delta FW4 Revision*	
	459	1113	*reserved*	
	45A	1114	*Max Solar 1 Voltage*	
	45B	1115	*Max Solar 2 Voltage*	
	45C	1116	*Max Solar 3 Voltage*	
	45D	1117	*Max Solar 4 Voltage*	
	45E	1118	*Max AC Voltage*	
	45F	1119	*Max Bus total Voltage*	
6	460	1120	*TestData0*	
	461	1121	*TestData1*	
	462	1122	*TestData2*	
	463	1123	*TestData3*	
	464	1124	*TestData4*	
	465	1125	*TestData5*	
	466	1126	*TestData6*	
	467	1127	*TestData7*	
	468	1128	*TestData8*	
	469	1129	*TestData9*	
	46A	1130	*TestDataA*	
	46B	1131	*TestDataB*	
	46C	1132	*reserved*	
	46D	1133	*reserved*	
	46E	1134	*reserved*	
	46F	1135	*reserved*	
7	470	1136	*Serial number 0,1*	
	471	1137	*Serial number 2,3*	
	472	1138	*Serial number 4,5*	
	473	1139	*Serial number 6,7*	
	474	1140	*Serial number 8,9*	
	475	1141	*Serial number 10,11*	
	476	1142	*Serial number 12,13*	
	477	1143	*Serial number 14,15*	
	478	1144	*reserved*	
	479	1145	*reserved*	
	47A	1146	*reserved*	
	47B	1147	*reserved*	
	47C	1148	*reserved*	

	47D	1149	*reserved*		
	47E	1150	*reserved*		
	47F	1151	*reserved*		
8	480	1152	*Max Temperature 1*	*unit : 1C*	TL5k:Max Amb Temp
	481	1153	*Max Temperature 2*	*unit : 1C*	TL5k:Max BoostHs Temp
	482	1154	*Max Temperature 3*	*unit : 1C*	TL5k:Max InvHs Temp
	483	1155	*Max Temperature 4*	*unit : 1C*	
	484	1156	*Max Temperature 5*	*unit : 1C*	
	485	1157	*Max Temperature 6*	*unit : 1C*	
	486	1158	*Max Temperature 7*	*unit : 1C*	
	487	1159	*Max Temperature 8*	*unit : 1C*	
	488	1160	*Solar RLeakage1*		TL5K:Solar 1 RLeakage1
	489	1161	*Solar RLeakage2*		TL5K:Solar 1 RLeakage2
	48A	1162	*Solar RLeakage3*		TL5K:Solar 1 RLeakage3
	48B	1163	*Solar RLeakage4*		TL5K:Solar 1 RLeakageAvg
	48C	1164	*Solar RLeakage5*		TL5K:Adins_adc
	48D	1165	*Solar RLeakage6*		
	48E	1166	*Solar RLeakage7*		
	48F	1167	*Solar RLeakage8*		
9	490	1168	*Installation Year*	*Ex:2000*	
	491	1169	*Installation Month*	*Ex:1*	
	492	1170	*Installation Date*	*Ex:1*	
	493	1171			
	494	1172			
	495	1173			
	496	1174			
	497	1175			
	498	1176			
	499	1177			
	49A	1178			
	49B	1179			
	49C	1180			
	49D	1181			
	49E	1182			
	49F	1183			
10	4A0	1184	*Event time (index+0)*	1184: Low word,1185:High Word.	
	4A1	1185		The cumulative seconds from 1970	
	4A2	1186	*Event time (index+1)*	0xFFFFFFFF: Not available	
	4A3	1187			
	4A4	1188	*Event time (index+2)*		
	4A5	1189			
	4A6	1190	*Event time (index+3)*		
	4A7	1191			
	4A8	1192	*Event time (index+4)*		

	4A9	1193	
	4AA	1194	*Event time (index+5)*
	4AB	1195	
	4AC	1196	*Event time (index+6)*
	4AD	1197	
	4AE	1198	*Event time (index+7)*
	4AF	1199	
11	4B0	1200	*Event time (index+8)*
	4B1	1201	
	4B2	1202	*Event time (index+9)*
	4B3	1203	
	4B4	1204	
	4B5	1205	
	4B6	1206	
	4B7	1207	
	4B8	1208	
	4B9	1209	
	4BA	1210	
	4BB	1211	
	4BC	1212	
	4BD	1213	
	4BE	1214	
	4BF	1215	
12	4C0	1216	
	4C1	1217	
	4C2	1218	
	4C3	1219	
	4C4	1220	
	4C5	1221	
	4C6	1222	
	4C7	1223	
	4C8	1224	
	4C9	1225	
	4CA	1226	
	4CB	1227	
	4CC	1228	
	4CD	1229	
	4CE	1230	
	4CF	1231	
	4D0	1232	
	4D1	1233	
	4D2	1234	
	4D3	1235	
	4D4	1236	

13	4D5	1237		
	4D6	1238		
	4D7	1239		
	4D8	1240		
	4D9	1241		
	4DA	1242		
	4DB	1243		
	4DC	1244		
	4DD	1245		
	4DE	1246		
	4DF	1247		
14	4E0	1248		
	4E1	1249		
	4E2	1250		
	4E3	1251		
	4E4	1252		
	4E5	1253		
	4E6	1254		
	4E7	1255		
	4E8	1256		
	4E9	1257		
	4EA	1258		
	4EB	1259		
	4EC	1260		
	4ED	1261		
	4EE	1262		
	4EF	1263		
15	4F0	1264	*Utility over voltage (OVR)*	0.1V
	4F1	1265	*Utility under voltage (UVR)*	0.1V
	4F2	1266	*OVR&UVR setting time*	0.01 sec.
	4F3	1267	*Utility over frequency(OFR) for 60Hz*	0.01Hz
	4F4	1268	*Utility under frequency(UFR) for 60Hz*	0.01Hz
	4F5	1269	*OFR&UFR setting time*	0.01 sec.
	4F6	1270	*Recover time*	1 sec.
	4F7	1271	*Suppression voltage*	0.1V
	4F8	1272	*Islanding passive setting*	0-off 1-on
	4F9	1273	*Suppression level*	%
	4FA	1274	*Utility over frequency(OFR) for 50Hz*	0.01Hz
	4FB	1275	*Utility over frequency(OFR) for 50Hz*	0.01Hz
	4FC	1276	*RCMU confirm time setting*	0-off 1-on
	4FD	1277	*In/Out door setting*	0-In door 1-Out door
	4FE	1278		
	4FF	1279		

1056	Measurement Index									
	0x00(AC Output	**0x01(AC Output**	**0x02(AC Output**	**0x10(AC Output**	**0x11(AC Output**	**0x12(AC Output**	**0x30(DC Input 1)**	**0x31(DC Input 2)**	**0x40(Bus Pos)**	**0x41(Bus Neg)**
1057	*Voltage(Vba)*	*Voltage(Vbc)*	*Voltage(Vca)*				*Voltage*	*Voltage*	*Voltage*	*Voltage*
1058	*Current*	*Current*	*Current*				*Current*	*Current*		
1059	*Wattage*	*Wattage*	*Wattage*				*Wattage*	*Wattage*		
1060	*Frequency*	*Frequency*	*Frequency*							
1061										
1062	*Redundant Voltage*	*Redundant Voltage*	*Redundant Voltage*	*Redundant Voltage*	*Redundant Voltage*	*Redundant Voltage*				
1063	*Redundant Frequency*	*Redundant Frequency*	*Redundant Frequency*							
1064										
1065										
1066										
1067										
1068	*Adc Voltage*	*Adc Voltage*	*Adc Voltage*				*Adc Voltage*	*Adc Voltage*	*Adc Voltage*	*Adc Voltage*
1069	*Adc Current*	*Adc Current*	*Adc Current*				*Adc Current*	*Adc Current*		
1070	*Adc Wattage*	*Adc Wattage*	*Adc Wattage*				*Adc Wattage*	*Adc Wattage*		
1071	*Adc Redundant Voltage*	*Adc Redundant Voltage*	*Adc Redundant Voltage*	*Adc Redundant Voltage*	*Adc Redundant Voltage*	*Adc Redundant Voltage*				

	Address (Hex)	Address (Dec)	Description	
0	800	2048	*Day-0 Wh*	*2048: Low word,2049:High Word* *unit : Wh*
	801	2049		
	802	2050	*Day-1 Wh*	*unit : Wh*
	803	2051		
	804	2052	*Day-2 Wh*	*unit : Wh*
	805	2053		
	806	2054	*Day-3 Wh*	*unit : Wh*
	807	2055		
	808	2056	*Day-4 Wh*	*unit : Wh*
	809	2057		
	80A	2058	*Day-5 Wh*	*unit : Wh*
	80B	2059		
	80C	2060	*Day-6 Wh*	*unit : Wh*
	80D	2061		
	80E	2062	*Day-7 Wh*	*unit : Wh*
	80F	2063		
1	810	2064	*Day-8 Wh*	*unit : Wh*
	811	2065		
	812	2066	*Day-9 Wh*	*unit : Wh*
	813	2067		
	814	2068	*Day-10 Wh*	*unit : Wh*
	815	2069		
	816	2070	*Day-11 Wh*	*unit : Wh*
	817	2071		
	818	2072	*Day-12 Wh*	*unit : Wh*
	819	2073		
	81A	2074	*Day-13 Wh*	*unit : Wh*
	81B	2075		
	81C	2076	*Day-14 Wh*	*unit : Wh*
	81D	2077		
	81E	2078	*Day-15 Wh*	*unit : Wh*
	81F	2079		
2	820	2080	*Day-16 Wh*	*unit : Wh*
	821	2081		
	822	2082	*Day-17 Wh*	*unit : Wh*
	823	2083		
	824	2084	*Day-18 Wh*	*unit : Wh*
	825	2085		
	826	2086	*Day-19 Wh*	*unit : Wh*
	827	2087		
	828	2088	*Day-20 Wh*	*unit : Wh*
	829	2089		
	82A	2090	*Day-21 Wh*	*unit : Wh*
	82B	2091		
	82C	2092	*Day-22 Wh*	*unit : Wh*
	82D	2093		
	82E	2094	*Day-23 Wh*	*unit : Wh*
	82F	2095		

3	830	2096	*Day-24 Wh*	unit : Wh
	831	2097		
	832	2098	*Day-25 Wh*	unit : Wh
	833	2099		
	834	2100	*Day-26 Wh*	unit : Wh
	835	2101		
	836	2102	*Day-27 Wh*	unit : Wh
	837	2103		
	838	2104	*Day-28 Wh*	unit : Wh
	839	2105		
	83A	2106	*Day-29 Wh*	unit : Wh
	83B	2107		
	83C	2108	*Day-30 Wh*	unit : Wh
	83D	2109		
	83E	2110	*Day-31 Wh*	unit : Wh
	83F	2111		
4	840	2112	*Month-0 Wh*	unit : Wh
	841	2113		
	842	2114	*Month--1 Wh*	unit : Wh
	843	2115		
	844	2116	*Month--2 Wh*	unit : Wh
	845	2117		
	846	2118	*Month--3 Wh*	unit : Wh
	847	2119		
	848	2120	*Month--4 Wh*	unit : Wh
	849	2121		
	84A	2122	*Month--5 Wh*	unit : Wh
	84B	2123		
	84C	2124	*Month--6 Wh*	unit : Wh
	84D	2125		
	84E	2126	*Month--7 Wh*	unit : Wh
	84F	2127		
5	850	2128	*Month--8 Wh*	unit : Wh
	851	2129		
	852	2130	*Month--9 Wh*	unit : Wh
	853	2131		
	854	2132	*Month--10 Wh*	unit : Wh
	855	2133		
	856	2134	*Month--11 Wh*	unit : Wh
	857	2135		
	858	2136	*Month--12 Wh*	unit : Wh
	859	2137		
	85A	2138	*Month--13 Wh*	unit : Wh
	85B	2139		
	85C	2140	*Month--14 Wh*	unit : Wh
	85D	2141		
	85E	2142	*Month--15 Wh*	unit : Wh
	85F	2143		
	860	2144	*Month-16 Wh*	unit : Wh

6	861	2145	*Month--16 Wh*	unit : Wh
	862	2146	*Month--17 Wh*	unit : Wh
	863	2147		
	864	2148	*Month--18 Wh*	unit : Wh
	865	2149		
	866	2150	*Month--19 Wh*	unit : Wh
	867	2151		
	868	2152	*Month--20 Wh*	unit : Wh
	869	2153		
	86A	2154	*Month--21 Wh*	unit : Wh
	86B	2155		
	86C	2156	*Month--22 Wh*	unit : Wh
	86D	2157		
	86E	2158	*Month--23 Wh*	unit : Wh
	86F	2159		
7	870	2160	*Quarter 0*	unit : Wh(average power of each 15min)
	871	2161	*Quarter 1*	unit : Wh
	872	2162	*Quarter 2*	unit : Wh
	873	2163	*Quarter 3*	unit : Wh
	874	2164	*Quarter 4*	unit : Wh
	875	2165	*Quarter 5*	unit : Wh
	876	2166	*Quarter 6*	unit : Wh
	877	2167	*Quarter 7*	unit : Wh
	878	2168	*Quarter 8*	unit : Wh
	879	2169	*Quarter 9*	unit : Wh
	87A	2170	*Quarter 10*	unit : Wh
	87B	2171	*Quarter 11*	unit : Wh
	87C	2172	*Quarter 12*	unit : Wh
	87D	2173	*Quarter 13*	unit : Wh
	87E	2174	*Quarter 14*	unit : Wh
	87F	2175	*Quarter 15*	unit : Wh
			~~	
12	8C0	2240	*Quarter 80*	unit : Wh
	8C1	2241	*Quarter 81*	unit : Wh
	8C2	2242	*Quarter 82*	unit : Wh
	8C3	2243	*Quarter 83*	unit : Wh
	8C4	2244	*Quarter 84*	unit : Wh
	8C5	2245	*Quarter 85*	unit : Wh
	8C6	2246	*Quarter 86*	unit : Wh
	8C7	2247	*Quarter 87*	unit : Wh
	8C8	2248	*Quarter 88*	unit : Wh
	8C9	2249	*Quarter 89*	unit : Wh
	8CA	2250	*Quarter 90*	unit : Wh
	8CB	2251	*Quarter 91*	unit : Wh
	8CC	2252	*Quarter 92*	unit : Wh
	8CD	2253	*Quarter 93*	unit : Wh
	8CE	2254	*Quarter 94*	unit : Wh
	8CF	2255	*Quarter 95*	unit : Wh

900	2304	*Day-0 Runtime*	*2304:Low Word, 2305: High Word unit : 1 second*
901	2305		
902	2306	*Day-1 Runtime*	
903	2307		
904	2308	*Day-2 Runtime*	
905	2309		
906	2310	*Day-3 Runtime*	
907	2311		
908	2312	*Day-4 Runtime*	
909	2313		
90A	2314	*Day-5 Runtime*	
90B	2315		
90C	2316	*Day-6 Runtime*	
90D	2317		
90E	2318	*Day-7 Runtime*	
90F	2319		
910	2320	*Day-8 Runtime*	
911	2321		
912	2322	*Day-9 Runtime*	
913	2323		
914	2324	*Day-10 Runtime*	
915	2325		
916	2326	*Day-11 Runtime*	
917	2327		
918	2328	*Day-12 Runtime*	
919	2329		
91A	2330	*Day-13 Runtime*	
91B	2331		
91C	2332	*Day-14 Runtime*	
91D	2333		
91E	2334	*Day-15 Runtime*	
91F	2335		
920	2336	*Day-16 Runtime*	
921	2337		
922	2338	*Day-17 Runtime*	
923	2339		
924	2340	*Day-18 Runtime*	
925	2341		
926	2342	*Day-19 Runtime*	
927	2343		
928	2344	*Day-20 Runtime*	
929	2345		
92A	2346	*Day-21 Runtime*	
92B	2347		
92C	2348	*Day-22 Runtime*	
92D	2349		
92E	2350	*Day-23 Runtime*	
92F	2351		
930	2352	*Day-24 Runtime*	

931	2353	*Day-24 Runtime*	
932	2354	*Day-25 Runtime*	
933	2355		
934	2356	*Day-26 Runtime*	
935	2357		
936	2358	*Day-27 Runtime*	
937	2359		
938	2360	*Day-28 Runtime*	
939	2361		
93A	2362	*Day-29 Runtime*	
93B	2363		
93C	2364	*Day-30 Runtime*	
93D	2365		
93E	2366	*Day-31 Runtime*	
93F	2367		

	Address (Dec)		
0	3072	*Current even - Error0*	E15~E00
	3073	*Current even - Error1*	E31~E16
	3074	*Current even - Error2*	E47~E32
	3075	*Current even - Error3*	.. .. ..
	3076	*Current even - Error4*	.. .. ..
	3077	*Current even - Error5*	.. .. ..
	3078	*Current even - Error6*	.. .. ..
	3079	*Current even - Error7*	.. .. ..
	3080	*Current even - Error8*	.. .. ..
	3081	*Current even - Error9*	.. .. ..
	3082	*Current even - Error10*	.. .. ..
	3083	*Current even - Error11*	.. .. ..
	3084	*Current even - Error12*	.. .. ..
	3085	*Current even - Error13*	.. .. ..
	3086	*Current even - Error14*	.. .. ..
	3087	*Current even - Error15*	.. .. ..
1	3088	*Current even - Warning0*	W15~W00
	3089	*Current even - Warning1*	W31~W16
	3090	*Current even - Warning2*	W47~W32
	3091	*Current even - Warning3*	.. .. ..
	3092	*Current even - Warning4*	.. .. ..
	3093	*Current even - Warning5*	.. .. ..
	3094	*Current even - Warning6*	.. .. ..
	3095	*Current even - Warning7*	.. .. ..
	3096	*Current even - Warning8*	.. .. ..
	3097	*Current even - Warning9*	.. .. ..
	3098	*Current even - Warning10*	.. .. ..
	3099	*Current even - Warning11*	.. .. ..
	3100	*Current even - Warning12*	.. .. ..
	3101	*Current even - Warning13*	.. .. ..
	3102	*Current even - Warning14*	.. .. ..
	3103	*Current even - Warning15*	.. .. ..
	3104	*Current even - Fault0*	F15~F00
	3105	*Current even - Fault1*	F31~F16
	3106	*Current even - Fault2*	F47~F32
	3107	*Current even - Fault3*	.. .. ..
	3108	*Current even - Fault4*	.. .. ..
	3109	*Current even - Fault5*	.. .. ..

ex : E15 E14 E13 ············E01 E00

0100 1000 0000 0010

(occur E14,E11 and E01)

E31 E14 E13 ···········E17 E16

2	**3110**	***Current even - Fault6***	. . . . . .
	3111	***Current even - Fault7***	. . . . . .
	3112	***Current even - Fault8***	. . . . . .
	3113	***Current even - Fault9***	. . . . . .
	3114	***Current even - Fault10***	. . . . . .
	3115	***Current even - Fault11***	. . . . . .
	3116	***Current even - Fault12***	. . . . . .
	3117	***Current even - Fault13***	. . . . . .
	3118	***Current even - Fault14***	. . . . . .
	3119	***Current even - Fault15***	. . . . . .

Group	Code	Message	description	Standard	Criterion
Error	E01	AC Freq High	Grid freqency out of range.	AC Freq High	1. Frequency calculated by Red. CPU ≧ OFR setting value 2. Frequency calculated by DSP ≧ OFR setting value 3. Italy Self-test reaches OFR setting value
	E02	AC Freq Low	Grid freqency out of range.	AC Freq Low	1. Frequency calculated by Red. CPU ≦ UFR setting value 2. Frequency calculated by DSP ≦ UFR setting value 3. Italy Self-test reaches UFR setting value
	E07	AC Quality	Disconnect because of bad grid quality.	Grid Quality	Continuously ride-through due to harmonic >8.5% and
	E08	HW Phase Seq	Hardware problem.	HW Connect Fail	Can't judge Grid sequence (RST or RTS)
	E09	No Grid	No grid.	No Grid	1. Both DSP and Red. CPU calculate Ua, Ub and Uc RMS < 20Vac 2. DSP calculates Uba or Ubc half-cycle > 50ms
	E10	AC Volt Low	Grid Voltage Low.	AC Volt Low	1. DSP or Red. calculates Ua_rms < UVR setting value 2. Italy Self-test reacheses UVR setting value
	E11	AC Volt High	Grid Voltage High.	AC Volt High	1. DSP or Red. calculates Ua_rms > OVR setting value 2. Italy Self-test reaches OVR setting value
	E13	SL AC Volt High	Grid Voltage High for 10 min.	AC Volt High	DSP or Red. calculates Ua_rms > OVR Slow setting value
	E15	AC Volt Low	Grid Voltage Low.	AC Volt Low	1. DSP or Red. calculates Ub_rms < UVR setting value 2. Italy Self-test reaches UVR setting value
	E16	AC Volt High	Grid Voltage High.	AC Volt High	1. DSP or Red. calculates Ua_rms > OVR setting value 2. Italy Self-test reaches OVR setting value
	E18	SL AC Volt High	Grid Voltage High for 10 min.	AC Volt High	DSP or Red. calculates Ub_rms > OVR Slow setting value
	E20	AC Volt Low	Grid Voltage Low.	AC Volt Low	1. DSP or Red. calculates Uc_rms < UVR setting value 2. Italy Self-test reaches UVR setting value
	E21	AC Volt High	Grid Voltage High.	AC Volt High	1. DSP or Red. calculates Ua_rms > OVR setting value 2. Italy Self-test reaches OVR setting value
	E23	SL AC Volt High	Grid Voltage High for 10 min.	AC Volt High	DSP or Red. calculates Uc_rms > OVR Slow setting value
	E30	Sol1 High	Solar input over voltage. Inverter may be damaged.	Solar1 High	Udc1 > 1000V (more than 0.1s)
	E31	Sol2 High	Solar input over voltage. Inverter may be damaged.	Solar2 High	Udc2 > 1000V (more than 0.1s)
	E34	Insulation	Leak current between PV array and grounding.	Insulation	Array impedance in either input < 1200 kohm
Warning	W01	Sol1 Low	Solar input under voltage.	Solar1 Low	Inverter has passed Dump load test, Udc1 < 180Vdc more than 10s
	W02	Sol2 Low	Solar input under voltage.	Solar2 Low	Inverter has passed Dump load test, Udc2 < 180Vdc more than 10s
	W11	HW Fan	Fan Failure. Check, clean or replace fan.	HW FAN	Fan locked or Fan fail during fans operation
	F01	HW DC Inject	Disconnect because of DC injection..	HW DC Injection	DC injection to Grid is over setting value
	F02	HW DC Inject	Disconnect because of DC injection.	HW DC Injection	DC injection to Grid is over setting value
	F03	HW DC Inject	Disconnect because of DC injection.	HW DC Injection	DC injection to Grid is over setting value
	F05	Temperature	Disconnect because of temperature too high.	Temperature High	Power de-rating down to < 5% due to high temperature
	F06	HW NTC1	Hardware problem.	HW NTC1 Fail	NTC1 (RTM1) is open (A/D<75) or short (A/D>4030)

F07	**Temperature**	Disconnect because of temperature too high.	**Temperature Low**	Calculated temperature < -20℃ from one of (NTC1, NTC2, NTC3 and NTC4)
F08	**HW NTC2**	Hardware problem.	**HW NTC2 Fail**	NTC2 (RTB1) is open (A/D<75) or short (A/D>4030)
F09	**HW NTC3**	Hardware problem.	**HW NTC3 Fail**	NTC3 (RTG1) is open (A/D<75) or short (A/D>4030)
F10	**HW NTC4**	Hardware problem.	**HW NTC4 Fail**	NTC4 (RTH1) is open (A/D<75) or short (A/D>4030)
F13	**HW Relay**	Hardware problem.	**Relay Test Open**	During On Grid, 1. Uinv_ba_rms without Uinv_ba_rms ±16V 2. Uinv_bc_rms without Uinv_bc_rms ±16V
F15	**HW DSP ADC1**	Hardware problem.	**HW DSP ADC1**	During Off Grid, have passed Dump load test, Ia, Ib, Ic, Uba or Ubc offset without 1.5 ±0.3V
F16	**HW DSP ADC2**	Hardware problem.	**HW DSP ADC2**	1. During Boost Off, Ubus > [Max(Udc1, Udc2) +20V] 2. Max(Udc1, Udc2) > 50V and Ubus < [Max(Udc1, Udc2) -20V]
F17	**HW DSP ADC3**	Hardware problem.	**HW DSP ADC3**	During Off Gird, have passed Dump load test, 1. Idc1 signal > 1.0V 2. Idc2 signal > 1.0V 3. Iboost1 signal > 0.8V 4. Iboost2 signal > 0.8V
F18	**HW RED ADC1**	Hardware problem.	**HW Red ADC1**	During Off Gird, have passed Dump load test, 1. Uba_offset signal without 1.5±0.2V 2. Ubc_offset signal without 1.5±0.2V 3. Uinv_ba_offset signal without 1.5±0.2V 4. Uinv_bc_offset signal without 1.5±0.2V
F19	**HW RED ADC2**	Hardware problem.	**HW Red ADC2**	During Off Gird, have passed Dump load test, 1. Idc_injection_phaseA signal without 1.5±0.4V 2. Idc_injection_phaseB signal without 1.5±0.4V 3. Idc_injection_phaseC signal without 1.5±0.4V
F20	**HW Efficiency**	Hardware problem.	**HW Efficiency**	When Pout > (20KW * 0.2 = 4kW), efficiency < 70% or > 130%
F22	**HW COMM2**	Hardware problem.	**HW COMM2**	COMM2(DSP <-> Red.) fail more than 20s
F23	**HW COMM1**	Hardware problem.	**HW COMM1**	COMM1(COMM <-> DSP) fail more than 11mins
F24	**Ground Leakage**	Disconnect because of ground leakage.	**Ground Current**	During On Grid, residual current: 1. (RMS > 150mA) > 150ms 2. (|Average value| > 125mA) > 5ms 3. [|Average value| > 20mA and RMS > 110mA] > 75ms 4. [|Average value| > 50mA and RMS > 50mA] > 25ms
~~**F25**~~	~~**Insulation**~~	~~Leak current between PV array and grounding.~~	**Insulation**	Array impedance in either input < 1200 kohm

Fault	F26	HW Int Wire	Hardware problem.	HW Connect Fail	During On Grid and if (Ia_rms > 15A or Ib_rms > 15A or Ic_rms > 15A) 1. Ia_rms < (Ib_rms)/3 or < (Ic_rms)/3 2. Ib_rms < (Ia_rms)/3 or < (Ic_rms)/3 3. Ic_rms < (Ia_rms)/3 or < (Ic_rms)/3
	F27	HW RCMU	Hardware problem.	RCMU Fail	During Off Grid, 1. Residual current without ±14.44mA 2. RCMU no response > 6s During On Grid, RCMU no response > 500ms
	F28	HW Relay Short	Hardware problem.	Relay Test Short	During Relay-test, whether short, 1. Uinv_ba_rms within Uba_rms ±15V 2. Uinv_bc_rms within Ubc_rms ±15V
	F29	HW Relay Open	Hardware problem.	Relay Test Open	During Relay-test, whether open, 1. Uinv_ba_rms without Uba_rms ±15V 2. Uinv_bc_rms without Ubc_rms ±15V
	F30	HW Bus unbalance	Hardware problem.	Bus Unbalance	|UbusP-UbusN| > 150V
	F31	HW Bus OVR Plus	Hardware problem.	HW Bus OVR	UbusP > 515V
	F33	HW Bus OVR Min	Hardware problem.	HW Bus OVR	UbusN > 515V
	F35	HW Bus OVR	Hardware problem.	HW Bus OVR	(UbusP + UbusN) > 1000V
	F36	AC TR OCR	Disconnect because of AC over current.	AC Current High	(Ia_rms > 60A) > 50ms
	F37	AC OCR	Disconnect because of AC over current.	AC Current High	(Ia_rms > 36A) > 5s
	F38	AC TR OCR	Disconnect because of AC over current.	AC Current High	(Ib_rms > 60A) > 50ms
	F39	AC OCR	Disconnect because of AC over current.	AC Current High	(Ib_rms > 36A) > 5s
	F40	AC TR OCR	Disconnect because of AC over current.	AC Current High	(Ic_rms > 60A) > 50ms
	F41	AC OCR	Disconnect because of AC over current.	AC Current High	(Ic_rms > 36A) > 5s
	F42	HW CTA	Hardware problem.	HW CT A Fail	During CT_test for positive current, Iouta_dc without [(1.5 + 6.4m) ~ (1.5 + 12.1m)V] During CT_test for negative current, Iouta_dc without [(1.5 -12.1m) ~ (1.5 -6.4m)V]
	F43	HW CTB	Hardware problem.	HW CT B Fail	During CT_test for positive current, Ioutb_dc without [(1.5 + 6.4m) ~ (1.5 + 12.1m)V] During CT_test for negative current, Ioutb_dc without [(1.5 -12.1m) ~ (1.5 -6.4m)V]
	F44	HW CTC	Hardware problem.	HW CT C Fail	During CT_test for positive current, Ioutc_dc without [(1.5 + 6.4m) ~ (1.5 + 12.1m)V] During CT_test for negative current, Ioutc_dc without [(1.5 -12.1m) ~ (1.5 -6.4m)V]

	F45	HW AC OCR	Hardware problem.	HW AC OCR	During On Grid, Hardware OOCP occur within 2s and integrated up to 20
	F50	HW ZC	Hardware problem.	HW ZC Fail	Ua_rms, Ub_rms and Uc_rms > 30V and Not No Grid, Interval of zero-crossing signal of Uba or Ubc ≧ 2s
	F60	Sol1 OCR	Disconnect because of solar over current.	DC Current High	During On Grid, (Idc1 > 36A) > 0.2s
	F61	Sol2 OCR	Disconnect because of solar over current.	DC Current High	During On Grid, (Idc2 > 36A) > 0.2s
	F70	Sol1 TR OCR	Disconnect because of solar over current.	DC Current High	During On Grid, (Idc1 > 45A) > 0.1s
	F71	Sol2 TR OCR	Disconnect because of solar over current.	DC Current High	During On Grid, (Idc2 > 45A) > 0.1s